



Sara Samy

curriculum vitae

Education

- 2025 - present **Mathematics M.Sc.**, *Technische Universität Berlin*, Berlin, Germany.
Supervised by Prof. Alexander I. Bobenko and Dr. Jan Techter.
- 2018 - 2025 **Technomathematics B.Sc.**, *Technische Universität Berlin*, Berlin, Germany.
Thesis: *Circular sections of quadrics*. Supervised by Prof. Alexander I. Bobenko and Dr. Jan Techter.
- 2009 - 2014 **Chemical Engineering**, *Faculty of Engineering, Alexandria University*, Alexandria, Egypt.
Area of Specialization: *Petrochemical Engineering*.
- 2007 - 2009 **Thanaweya Amma**, *École Saint Vincent*, Alexandria, Egypt.

Research Experience

- 06.2025 - present **Research assistant**, *Computational Diagnosis and Therapy Planning*, Zuse Institute Berlin (ZIB), Germany.
- Developing topology-aware methods for non-rigid registration and tomographic reconstruction in medical imaging.
- 06.2024 - 08.2024 **SGI Fellow**, *Summer Geometry Initiative*, Massachusetts Institute of Technology, U.S.
An intensive summer research program in geometry processing.
- Developed output-sensitive collision detection for deformable models.
 - Implemented a global intersection analysis technique to resolve self-collision in cloth simulations.
 - Implemented an automatic sketch colorization method using winding number features on tori.
 - Analyzed algorithms for constructing solid surfaces from signed distance data (SDFs).
- 01.2021 - 06.2024 **Research assistant**, *Discretization in Geometry and Dynamics*, *SFB Transregio 109*, Technische Universität Berlin, Germany.
- Contributed to the development of *pyddg*, a python library for discrete differential geometry.
 - Produced mathematical visualizations for the film *Solving the Bonnet problem* by Ekaterina Ee and Prof. Alexander I. Bobenko.
 - Led a team of students to create visualizations for *Geometry I-II* module taught yearly at TU Berlin.

Previous Work

- 2010 - 2015 **Visual artist**
Selected exhibitions:
- Optimized textures (2015), Kestner Gesellschaft Hannover, Germany.
 - Cairo and Berlin (2015), Joint project of Goethe-Institut & German Embassy in Cairo, Egypt.
 - Was tust du, Objekt? (2015), Gypsum Gallery, Cairo, Egypt.
 - Profit | Prophet (2014), St. Johannes Church, Berlin Art Week, Germany.
 - New Aesthetics (2013), National Center for Contemporary Art, Yekaterinburg, Russia.
 - DVD Dead Drop (2013), Museum of the Moving Image, New York City, U.S.

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03.2012 - 08.2012 **Artist assistant**, *dOCUMENTA (13)*, Kassel, Germany.

- Provided administrative, technical and installation assistance to various artists during the quinquennial contemporary art exhibition Documenta 13.

05.2010 - 03.2012 **Newspaper Editor & Photographer**, *Akhbar el-Yom*, Alexandria, Egypt.

- Edited grammatical, stylistic, and content errors of article submissions in Arabic.
- Gathered and verified information through interviews, research and photography.

Languages

Arabic Mother tongue
English Fluent
German C1 (CEFR)

Technical Skills

Programming

Python, Julia, TypeScript, Rust, C++

Machine Learning Frameworks

Pytorch, JAX

Software

Blender, Houdini, Adobe Creative Suite, Zbrush

Tools

nix, vi/vim, L^AT_EX

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